

Pravesh Pandey

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PROFESSIONAL SUMMARY

Software Development Engineer with 3 years of experience architecting and delivering scalable, high-throughput distributed systems for Tier-1 services at Alexa. Strong background in Java, AWS, microservices, event-driven architectures, performance engineering, and production reliability. Built a deterministic NLU path handling 6K TPS and reducing latency by 400–550 ms. Led the development of LangForge, a translation and evaluation service adopted by 60+ Alexa teams.

EXPERIENCE

Amazon

Bengaluru, India

Software Development Engineer

May 2024 – Present

- Authored the low-level design and built a deterministic fast path that delivered Classic Alexa signals to Alexa+ without LLM inference, using an asynchronous publish-subscribe orchestrator and a 61 GB distributed cache fleet. Launched across 11 locales at 6K TPS, reducing latency by 400–550 ms and improving quality by up to 140 basis points.
- Architected locale-specific finite-state transducers (FSTs) to support Pattern Match hotfixes across Classic Alexa and Alexa+ in 18 locales; reduced artifact size by 58% and latency from 40 ms to 25 ms.
- Root-caused production incidents across five services, including FST corruption in a service handling 40K TPS; restored availability during a multi-Region outage and shipped alarms plus a five-minute mitigation runbook to close detection and response gaps.
- Designed and launched a five-agent orchestrator and production Model Context Protocol (MCP) server that automated the Operational Readiness Review (ORR) workflow, with nine safety controls and production observability. Helped complete an ORR in 11 agent-hours versus months of manual effort, driving cross-team adoption.
- Owned end-to-end development of LangForge, an automated translation and evaluation service, from a one-week proof of concept to adoption by 60+ Alexa teams. Integrated three partner systems and resolved 15+ edge cases, reducing onboarded-team CET failures from roughly 60% to near zero, shortening evaluation runtime from 18 hours to 12 minutes, and saving 160–200 engineering hours per cycle.

nFolks Data Solutions

Remote, India

Software Engineer

Oct 2023 – Apr 2024

- Contributed to a multithreaded Python data-processing framework that parallelized database reads and improved retrieval speed by 68%; optimized pandas pipelines and delivered Flask and SQL-based REST APIs through Jenkins CI/CD.

Phyt Health

Pune, India

Software Development Intern

Jan 2022 – Mar 2022

- Contributed to the development of an AI-guided fitness application using React, Node.js, TensorFlow, and C++; delivered real-time posture correction and exercise tracking to 500+ pilot users, earning average feedback of 4.2/5.

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript, TypeScript, C++, C#

Cloud & DevOps: Amazon Web Services (AWS: ECS/Fargate, Lambda, S3, DynamoDB, CloudWatch, KMS, IAM, CDK/CloudFormation, Step Functions, Kinesis, ElastiCache, ALB, Auto Scaling), Docker, Jenkins, Git, CI/CD

Backend & Data: Distributed systems, microservices, REST APIs, event-driven and publish-subscribe architecture, multithreading, high availability, fault tolerance, distributed caching, MySQL, MongoDB, Redis

Reliability & AI: System design, performance and capacity engineering, load/stress/resilience testing, observability, monitoring, incident response, root cause analysis (RCA), generative AI, large language models (LLMs), multi-agent systems, Model Context Protocol (MCP), deterministic NLU

EDUCATION & AWARDS

Motilal Nehru National Institute of Technology: Bachelor of Technology, 2019–2023

Awards

- Amazon Alexa AI AIDo Frugal Frontrunner (2025)
- Amazon Extra Mile (2025)
- MNNIT Project Exhibition Winner (2023)